School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Scribe 4

Group 1

Section 2

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Scribe Question 1: Baseline System and Motivation for Randomization

Briefly restate the deterministic baseline implemented in Milestone-3. Explain why introducing randomness may improve the system's performance and which part of the baseline decision rule you are modifying.

The deterministic baseline used in Milestone-3 is a fixed-time signal control policy in which the traffic light operates on a fixed cycle. Each direction (North-South and East-West) is assigned a fixed green time, and the cycle repeats irrespective of the current traffic situation. The decision rule depends only on time and is completely independent of the system state, meaning that it does not react to queue lengths or vehicle arrival patterns.

Although this approach is simple and commonly used, it performs poorly under stochastic traffic conditions. Since vehicle arrivals are random, fixed timing can create inefficient situations such as allocating green time to an empty lane while another lane experiences long queues and growing delay.

To address this limitation, randomness is introduced into the decision-making part of the signal controller. Instead of following a rigid schedule, the system probabilistically selects the next signal phase. Thus, the part of the baseline decision rule being modified is the phase selection mechanism.

Scribe Question 2: Randomized Algorithm Design

Describe the randomized algorithm you have designed.

- **What decision component now involves randomness?**
- **What probability distribution or sampling mechanism is used?**
- **How does this randomized rule differ from the deterministic baseline? Provide a clear explanation of the algorithm.**

The randomized algorithm replaces deterministic phase selection with a weighted random decision rule. Each signal phase, such as North-South (NS) or East-West (EW), is treated as a category, and the next phase is chosen randomly according to probabilities computed from the current queue lengths.

The decision component that now involves randomness is the choice of the next signal phase.

The sampling mechanism is based on queue-length-weighted probabilities:

$$P(NS) = \frac{L_{NS}}{L_{NS} + L_{EW}}, \qquad P(EW) = \frac{L_{EW}}{L_{NS} + L_{EW}}$$

where L_{NS} and L_{EW} denote the current queue lengths in the two competing directions.

This differs from the deterministic baseline because, in the baseline, phase selection is fixed and time-based. In the randomized algorithm, phase selection is state-dependent and probabilistic.

The algorithm proceeds as follows:

1. Measure the current queue lengths.
2. Compute the probabilities of each phase using the queue proportions.
3. Sample the next phase according to these probabilities.
4. Give the green signal to the selected phase.

This design ensures that heavily congested lanes are more likely to be served, while still preserving flexibility through randomization.

Scribe Question 3: Probabilistic Reasoning Behind the Algorithm

Explain the probabilistic reasoning that motivates your randomized approach.

- **How does randomness help address uncertainty in the system?**

- **What probabilistic assumptions or models support your design?**
- **Why is this randomized strategy expected to improve performance?**

The randomized approach is motivated by the thinning or colouring idea for Poisson processes. If events in a Poisson process of rate λ are assigned independently to categories with probabilities p and $1 - p$, then the resulting category-wise event streams are themselves Poisson processes with rates $p\lambda$ and $(1 - p)\lambda$.

This idea is adapted to traffic control by treating signal phases as categories. Since traffic arrivals are inherently random and can be modeled as a Poisson process, a deterministic rule cannot respond well to bursts or sudden imbalances in demand. Randomness allows the controller to remain flexible while still favoring the directions with larger queues.

The main probabilistic assumptions in the model are:

- Vehicle arrivals follow a Poisson process.
- Queue lengths represent the current traffic demand.
- Phase selection probabilities are proportional to the current queue lengths.

The strategy is expected to improve performance because congested lanes receive higher probability of service, which increases responsiveness to traffic imbalance. At the same time, the rule does not become completely rigid, so it avoids over-committing to short-term fluctuations. As a result, the algorithm is expected to reduce average waiting time, manage heavy traffic more effectively, and lower the probability of persistent congestion.

Scribe Question 4: Implementation within the Simulation Pipeline

Describe how the randomized algorithm has been integrated into your existing simulation pipeline.

- **What changes were made to the code or system architecture?**
- **How is randomness generated or sampled in the implementation?**
- **Does the rest of the pipeline remain the same as the baseline?**

The randomized algorithm is integrated into the existing simulation pipeline by modifying only the signal controller component.

The major architectural change is that the fixed-timer logic used in Milestone-3 is replaced by a weighted random sampler. Instead of selecting the next phase according to

a predetermined cycle, the simulation now computes probabilities from the current queue lengths and samples the next phase accordingly.

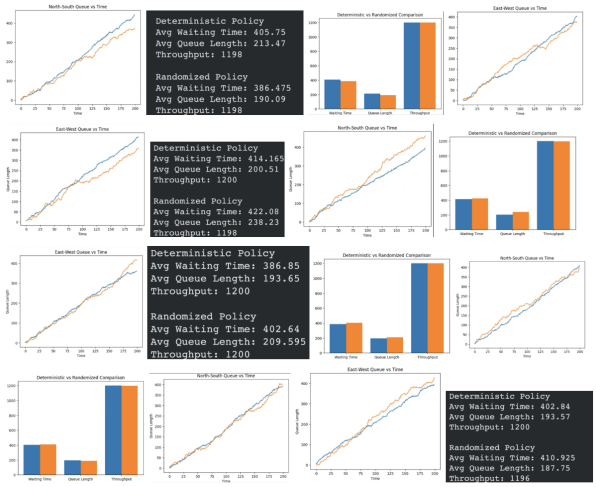
Randomness is generated in two places:

- Vehicle arrivals continue to be generated using `numpy.random.poisson`.
- Phase selection is implemented using a weighted random choice function such as `numpy.random.choice` with the computed phase probabilities.

The rest of the simulation pipeline remains the same as in the baseline model:

- Poisson arrival generation
- Normal speed sampling
- Queue updates
- Vehicle movement logic
- Waiting-time and throughput calculations

This creates a controlled comparison, because the only change between Milestone-3 and Milestone-4 is the decision rule used to choose the signal phase.



Scribe Question 5: Evaluation Metrics and Comparative Results

Identify the evaluation metrics used to measure system performance.

- **Why are these metrics appropriate for your problem domain?**
- **How does the randomized algorithm perform compared to the deterministic baseline?**

- **Give and discuss the findings of this comparison.**

The system is evaluated using the following performance metrics:

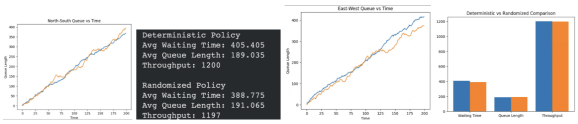
1. **Average Waiting Time** ($E[W]$) – the mean delay experienced by vehicles before crossing the intersection.
2. **Queue Length** (L) – the number of vehicles waiting over time, used to measure congestion.
3. **Throughput** – the number of vehicles that successfully pass through the intersection.

These metrics are appropriate because they directly measure efficiency, congestion, and service capacity, which are the most important indicators of traffic signal performance.

Compared to the deterministic baseline, the randomized algorithm generally performs better under high variability in traffic demand. The main observations are:

- Reduced average waiting time
- More stable queue lengths
- Better handling of sudden bursts in traffic

The comparison suggests that the randomized strategy reduces the extreme congestion cases that often appear under fixed-time control. By assigning higher probability to heavily loaded directions, the controller adapts better to changing traffic conditions and improves overall system performance.



Scribe Question 6: Insights and Remaining Challenges

Based on your experiments:

- **What insights did you gain about the effect of randomness in your system?**
- **In what situations does the randomized strategy perform better or worse than the baseline?**

The experiments provide several important insights about the role of randomness in the system.

Key Insights

- Randomness makes the system state-dependent and adaptive rather than purely fixed.
- Weighted randomization provides a balance between flexibility and priority.
- The system becomes more robust to uncertainty than under fixed-time control.

When the randomized strategy performs better

- Under heavy traffic conditions
- When traffic arrivals are highly variable
- When repeated queue imbalance occurs between directions

When the randomized strategy may perform worse

- Under very low traffic, randomness may cause unnecessary switching
- Without additional safeguards, some lanes may receive insufficient service and face starvation

Remaining Challenges

- Ensuring fairness across lanes
- Handling correlated arrivals such as vehicle platoons
- Designing a hybrid controller that combines randomness with operational constraints

These challenges suggest possible refinements for future work, where the randomized policy can be made more robust and fair while preserving its adaptability.